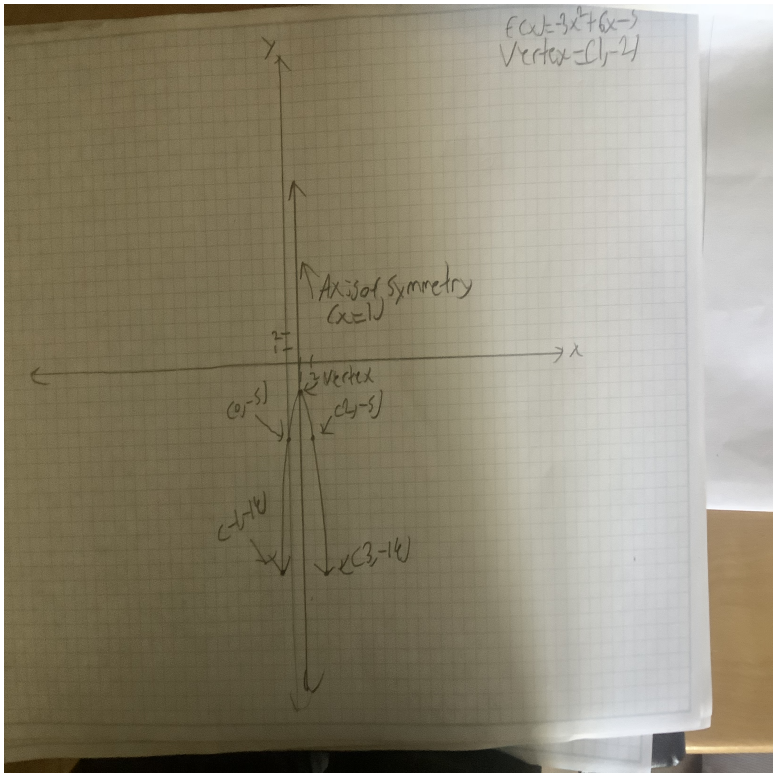
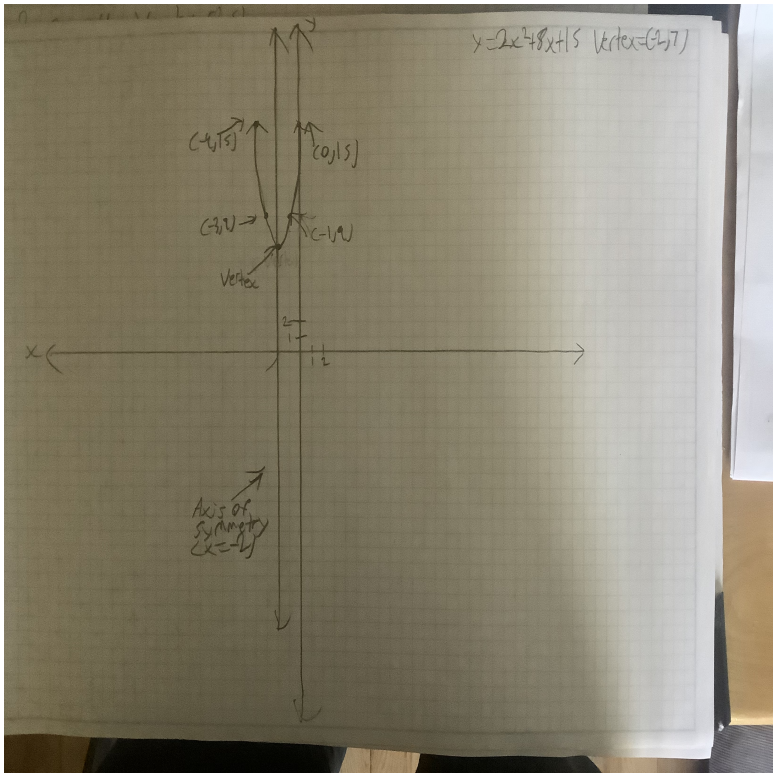
Quiz Exercises

September 20, 2023

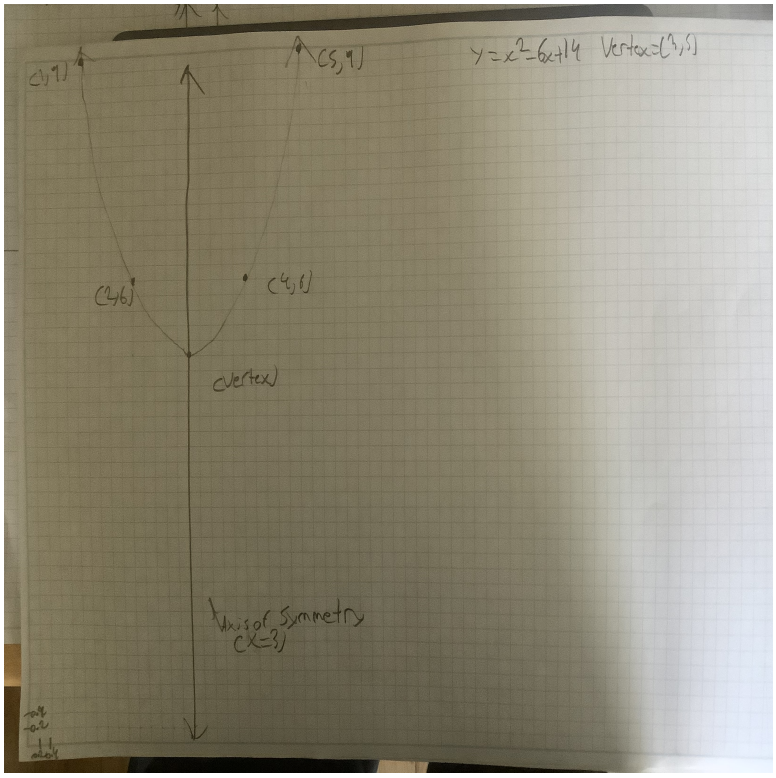
1



2


y=2x²+8x+15 Vertex=(-2,7)
y
x
(-4,15)
(0,15)
(-3,9)
(-1,9)
Vertex
2
1
1/2
Axis of Symmetry (x=-2)

3



4

$y = (x - 4)(x - 8)$
$\rightarrow = x^2 - 12x + 32$

5

$g(x) = -2(x + 3)(x - 7)$
$\rightarrow = -2(x^2 - 4x - 21)$
$\rightarrow = -2x^2 + 8x + 42$

6

$y = 5(x+6)^2 - 2$
$\rightarrow = 5(x^2 + 12x + 36) - 2$
$\rightarrow = 5x^2 + 60x + 180$
$\rightarrow = 5x^2 + 60x + 178$

7

$x^2 + 9x + 20 = 0$
$\rightarrow (x+5)(x+4) = 0$
$\rightarrow x = -5 \vee x = -4$

8

$n^2 - 11n + 24 = 0$
$\rightarrow (n-8)(n-3) = 0$
$\rightarrow n = 8 \vee n = 3$

9

$z^2 - 3z - 40 = 0$
$\rightarrow (z-8)(z+5) = 0$
$\rightarrow z = 8 \vee z = -5$

10

$5s^2 - 14s - 3 = 0$
$\rightarrow (5s+1)(s-3) = 0$
$\rightarrow s = -\frac{1}{5} \vee s = 3$

11

$7a^2 - 30a + 8 = 0$
$\rightarrow (7a - 2)(a - 4) = 0$
$\rightarrow a = \frac{2}{7} \vee a = 4$

12

$4x^2 + 20x + 25 = 0$
$\rightarrow (2x)^2 + 20x + 5^2 = 0$
$\rightarrow (2x + 5)^2 = 0$
$\rightarrow x = -\frac{5}{2}$

13

$R(x) = (140 - 10x)(50 + 5x)$ First, we find the x-coordinate of the vertex, by changing this equation into intercept form:
$-50(x - 14)(x + 10)$, and thus, the x coordinate is $\frac{14-10}{2} = 2$, meaning the maximum value is $(140 - 10(2))(50 + 5(2)) = 7200$. Hence, the store should charge $140 - 10(2) = \$120$ in order to make a maximum profit of \$7200.